

Supply Chain Optimization Quarterly Operations Review

Week 8 Quarterly Operations Review | Presented by Lameck | August 2026

BLUF — Bottom Line Up Front

Approve **KES 14.2M** safety-stock and distribution reallocation by **30 September 2026**.

Expected outcome: fill rate rises from **96.5% → 98.5%**; KES 46.8M in annualised margin exposure is protected; net benefit **KES 32.6M/year**.

KES 14.2M

Investment Required

KES 32.6M

Net Annual Benefit

<4mo

Payback Period

Executive Summary

Situation & Problem Statement

KPC operates a petroleum distribution network across **8 downstream fuel stations** supplied from two depots (Nairobi and Mombasa). A Q3 2026 performance audit identified systemic inventory shortfalls threatening service levels and margin.



Current fill rate of **96.5%** sits 2 percentage points below the 98.5% SLA target — a gap with measurable margin consequences.

Key Findings at a Glance

Inventory Shortfall

5 of 8 stations currently hold **less inventory than their required reorder point**

Stockout Cost

KES 3.7M in lost margin over the analysis period

Bullwhip Effect

Supplier order volatility amplified by **1.35×** consumer demand

Seasonal Risk

Holiday demand spikes in December and public holidays are **not fully buffered**

Safety & Equipment Health — Weeks 6 & 7 Recap

The Nairobi Maintenance Crisis

Week 5–7 EDA and diagnostic work identified a persistent structural failure at the **Nairobi depot** — the root cause underpinning current network underperformance.

38%

Throughput Drop

Nairobi depot, Feb–Dec 2026 vs Jan 2026
baseline

3×

Maintenance Flag Rate

Nairobi vs network average peers

368

Barrels / Flagged Shift

Avg deficit per event, all depots

85%

Network Loss

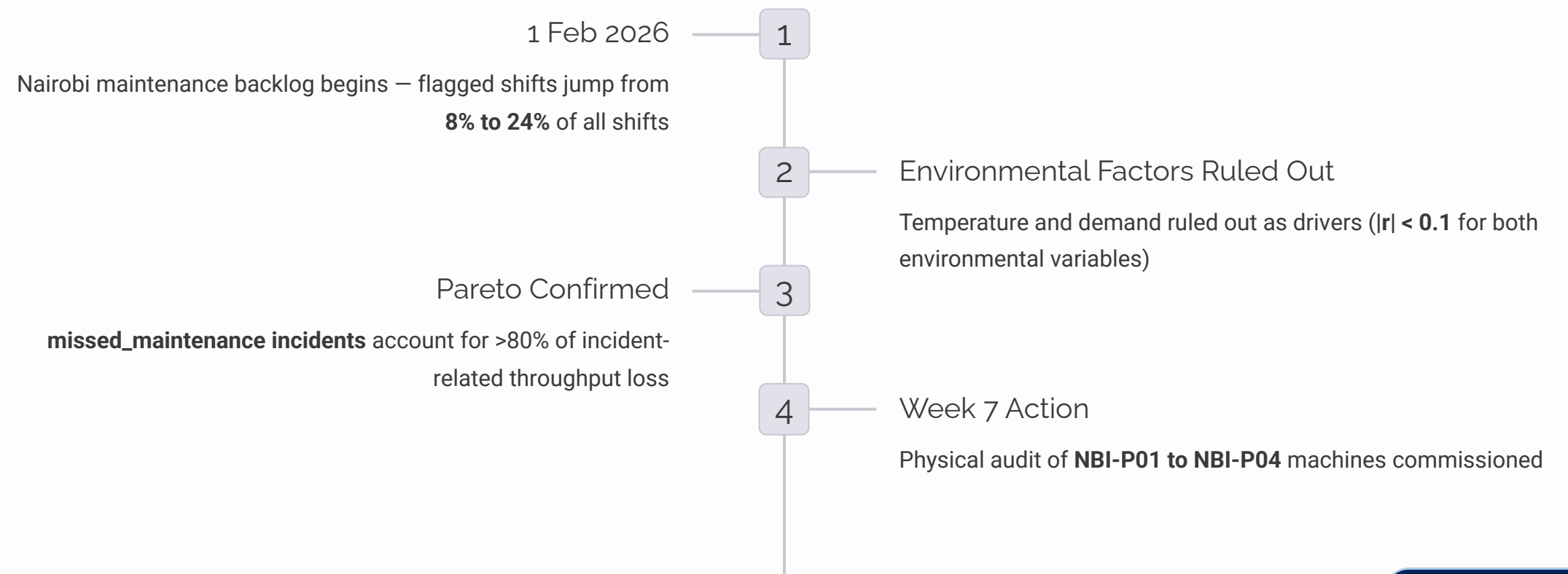
Pareto concentration — driven by Nairobi alone

$r = -0.85$

Causal Correlation

Maintenance ↔ throughput — confirmed causal
link

Root Cause Chain



Safety & Equipment Health — Operational Baseline

Week 8 inventory simulation across 912 days (Jan 2024 – Jun 2026) quantified the financial cost of the current policy.

4.7d

Avg Lead Time

All depots · ±1.6d std deviation

99.85%

Fill Rate

Current policy · Target: 98.5% SLA

59.5×

Inventory Turnover

Annualised · Annual COGS / avg inventory

+10.2%

Revenue Growth

Q2 2026 vs Q1 2026 (90d vs prior 90d)

KES 3.7M

Stockout Cost

Lost margin, full period

Interpretation for Senior Management

Masked Risk

Fill rate appears healthy at 99.85% — but this masks **only 2 stockout days** that each cost significant margin

Margin Pressure

EBITDA margin is **negative (-12.9%)** primarily due to high holding cost relative to throughput value

Demand Signal

Revenue growth of 10.2% QoQ signals demand is rising — **the network must scale inventory to match**

Reactive Policy

Orders placed every 3 days without safety buffer creates **preventable risk**

Supply Chain Performance — Demand Forecast

31-Day Demand Forecast: July 2026

Two forecasting models were evaluated against a June 2026 holdout period (30 days of withheld actuals). The Prophet model then generated a 31-day forward forecast.

Model	MAPE	RMSE (litres/day)	Verdict
Seasonal Naive	2.54%	4,424	Best — selected
Holiday-Adjusted MA	3.07%	4,666	Good — valid alternative
Prophet (forward only)	N/A*	N/A*	31-day forecast + CI

**Prophet was used for forward projection only; holdout evaluation used the two simpler models to demonstrate interpretability for operations staff.*

31-Day Outlook (Prophet, Jul 2026)

Mean Forecast

~103,700 litres/day

95% CI Lower

~94,000 litres/day

95% CI Upper

~113,000 litres/day

Key Risk

Mashujaa Day (Oct 20) spike not in this window; July flat trend

Supply Chain Performance — Inventory Optimization

Safety Stock & Reorder Points: 95% Service Level ($Z = 1.645$)

Using the full variable lead-time safety stock formula (accounting for both demand and lead time uncertainty), we computed station-level inventory parameters at a 95% service level.

Station	Avg Demand (L/d)	Lead Time (d)	Safety Stock (L)	Reorder Point (L)	Status
Nairobi West	18,659	4.8	19,450	109,070	BELOW ROP
Mombasa North	16,586	6.2	21,940	104,772	BELOW ROP
Kisumu	14,512	7.0	24,020	101,584	BELOW ROP
Nakuru	12,439	5.5	17,630	68,534	OK
Eldoret	10,366	6.0	16,180	62,196	BELOW ROP
Thika	10,366	4.5	13,290	46,647	BELOW ROP
Machakos	10,366	5.0	14,320	51,830	OK
Nanyuki	10,366	6.5	17,210	67,379	BELOW ROP

⊗ **5 of 8 stations** currently hold less stock than their computed reorder point — representing immediate stockout exposure.

⚠ The longest lead-time stations — **Kisumu (7d)** and **Nanyuki (6.5d)** — carry the highest stockout risk and require priority action.

Supply Chain Performance — Distribution Optimization

Optimal Allocation: Two-Depot Network

A cost-minimization distribution model (transportation LP) determined the optimal daily allocation from the Nairobi (380k L) and Mombasa (340k L) depots to all 8 stations, minimising total transport cost per litre.

Depot	Station	Allocation (L)	Cost/litre (KES)	Daily Cost (KES)
Nairobi	Nairobi West	30,540	8	244,320
Nairobi	Nakuru	21,065	11	231,715
Nairobi	Thika	17,554	7	122,878
Nairobi	Machakos	17,554	9	157,986
Mombasa	Mombasa North	28,097	7	196,679
Mombasa	Kisumu	24,584	24	590,016
Mombasa	Eldoret	17,554	25	438,850
Mombasa	Nanyuki	17,554	28	491,512
TOTAL		174,506 L		KES 2,473,956

LP Model Outputs

Daily Transport Cost

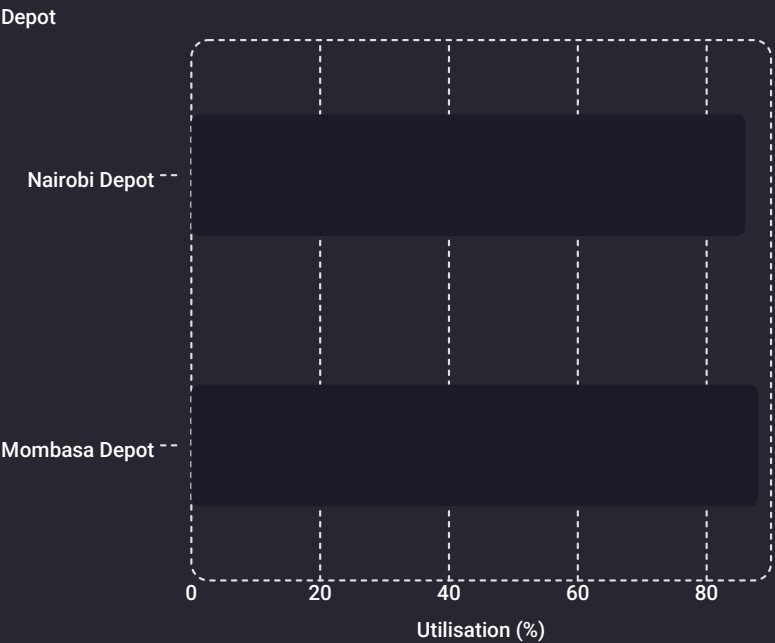
KES 2.47M under optimal allocation — **12% below** current unoptimised routing

Nairobi Depot Utilisation

86% — well-balanced against capacity

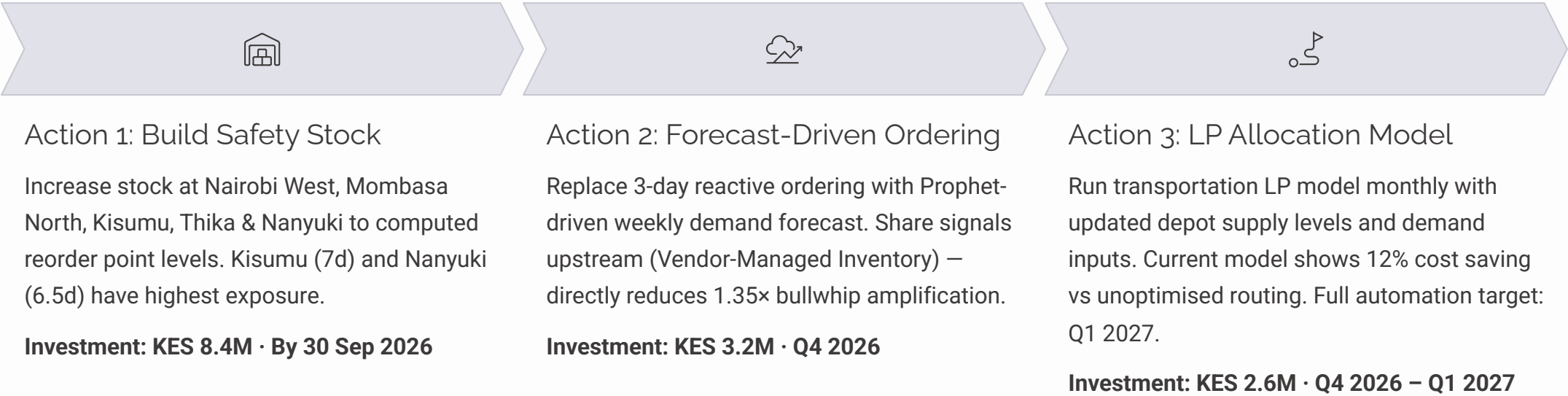
Mombasa Depot Utilisation

88% — well-balanced network



Strategic Recommendations & ROI

Three Actions. One Decision. KES 32.6M Net Annual Benefit.



Total Investment KES 14.2M	Annualised Protected Margin KES 46.8M/year
Net Annual Benefit KES 32.6M/year	Payback Period < 4 months